

Assignment 2
Data Acquisition and Survey Methods
Survey Analysis Report

Group 39

2026

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Introduction

This report analyzes the relationship between meal habits and perceived study productivity among university students. The focus is on three research questions related to breakfast frequency, the number of main meals per day, and self-reported study productivity.

The report addresses the following three research questions and hypotheses:

1. **RQ1:** Is the frequency of eating breakfast associated with students' perceived study productivity?
H1: Students who eat breakfast more often report higher perceived study productivity.
2. **RQ2:** Is the number of main meals per day associated with students' perceived study productivity?
H2: Students who eat more main meals per day report higher perceived study productivity.
3. **RQ3:** Is the frequency of eating breakfast associated with the number of main meals students usually eat per day?
H3: Students who eat breakfast more often also report eating more main meals per day.

For the analysis, only the variables relevant to these research questions were selected from the full survey dataset. These include three demographic variables and the three Group 39 survey variables on breakfast frequency, main meals per day, and perceived study productivity.

Data Preparation

```
# Load the survey data and keep the original column names
data <- read.csv("../data/Survey_results_2026.csv", check.names = FALSE)

# Select only the columns relevant for our group
group39_data <- data[, c(
  "DQ How old are you?",
  "DQ What is your gender?",
  "DQ What is your academic program?",
  "G39 How often do you eat breakfast during a typical study week?",
  "G39 How many main meals do you usually eat per day, excluding snacks?",
  "G39 How productive do you usually feel during your study sessions?"
)]

# Rename the columns to shorter names
names(group39_data) <- c(
  "age",
  "gender",
  "academic_program",
  "breakfast_frequency",
  "main_meals_per_day",
  "study_productivity"
)

# Convert breakfast frequency to an ordered factor
group39_data <- group39_data %>%
  mutate(
    main_meals_per_day = as.numeric(main_meals_per_day),
    study_productivity = as.numeric(study_productivity),
```

```

breakfast_frequency = factor(
  breakfast_frequency,
  levels = c(
    "Never",
    "1-2 days per week",
    "3-4 days per week",
    "5-6 days per week",
    "Every day"
  ),
  ordered = TRUE
),
breakfast_score = as.numeric(breakfast_frequency) - 1
)

# Prepare demographic variables for the additional exploratory analysis
group39_data <- group39_data %>%
  mutate(
    age = factor(
      age,
      levels = c("18-20", "21-23", "23-25", "25-30", "30-35", "over 35"),
      ordered = TRUE
    ),
    gender = factor(gender),
    academic_program = factor(academic_program)
  )

# Show the first rows
head(group39_data)

```

##	age	gender	academic_program	breakfast_frequency	main_meals_per_day
## 1	25-30	Male	Business Informatics	Every day	3
## 2	21-23	Male	Data Science	3-4 days per week	2
## 3	21-23	Female	Data Science	Every day	3
## 4	25-30	Female	Data Science	Every day	3
## 5	over 35	Female	Data Science	Never	2
## 6	25-30	Male	Data Science	1-2 days per week	1

##	study_productivity	breakfast_score
## 1	3	4
## 2	3	2
## 3	3	4
## 4	4	4
## 5	3	0
## 6	5	1

The selected dataset contains demographic variables as well as the three variables needed to test the research hypotheses. The variables were converted into appropriate formats for the analysis. The number of main meals per day and study productivity were transformed into numeric variables. Breakfast frequency was converted into an ordered factor because the response categories follow a natural order from “Never” to “Every day”. Additionally, a numeric breakfast score was created to use breakfast frequency in the statistical correlation tests.

```
str(group39_data)
```

```
## 'data.frame':    209 obs. of  7 variables:
## $ age           : Ord.factor w/ 6 levels "18-20"<"21-23"<...: 4 2 2 4 6 4 4 4 5 4 ...
## $ gender        : Factor w/ 2 levels "Female","Male": 2 2 1 1 1 2 2 2 1 ...
## $ academic_program : Factor w/ 2 levels "Business Informatics",...: 1 2 2 2 2 2 2 2 2 ...
## $ breakfast_frequency: Ord.factor w/ 5 levels "Never"<"1-2 days per week"<...: 5 3 5 5 1 2 3 2 3 2 .
## $ main_meals_per_day : num  3 2 3 3 2 1 2 3 3 2 ...
## $ study_productivity : num  3 3 3 4 3 5 3 3 4 3 ...
## $ breakfast_score    : num  4 2 4 4 0 1 2 1 2 1 ...
```

```
colSums(is.na(group39_data))
```

```
##           age           gender  academic_program breakfast_frequency
##           0             0             0             0
## main_meals_per_day study_productivity  breakfast_score
##           0             0             0
```

The dataset contains 209 observations and 7 variables after data preparation. No missing values were found in any of the selected variables, so no observations had to be removed before the analysis. The following sections analyze each research question separately. For each hypothesis, we first use an exploratory visualization, then compute descriptive summary statistics, and finally apply an appropriate statistical test.

Hypothesis 1: Breakfast Frequency and Study Productivity

Research Question: Is the frequency of eating breakfast associated with students' perceived study productivity?

Hypothesis: Students who eat breakfast more often report higher perceived study productivity.

To test this hypothesis, we examine the relationship between two variables: **breakfast_frequency** and **study_productivity**. The variable **breakfast_frequency** is ordinal because its response categories range from “Never” to “Every day” and follow a meaningful order. The variable **study_productivity** is also ordinal, as it is measured on a 1-5 Likert scale, where higher values indicate higher perceived study productivity.

Exploratory Data Analysis

For the exploratory data analysis, we calculated the average perceived study productivity for each breakfast-frequency category and visualized it with a line chart. This visualization helps us identify potential trends in study productivity across different breakfast habits.

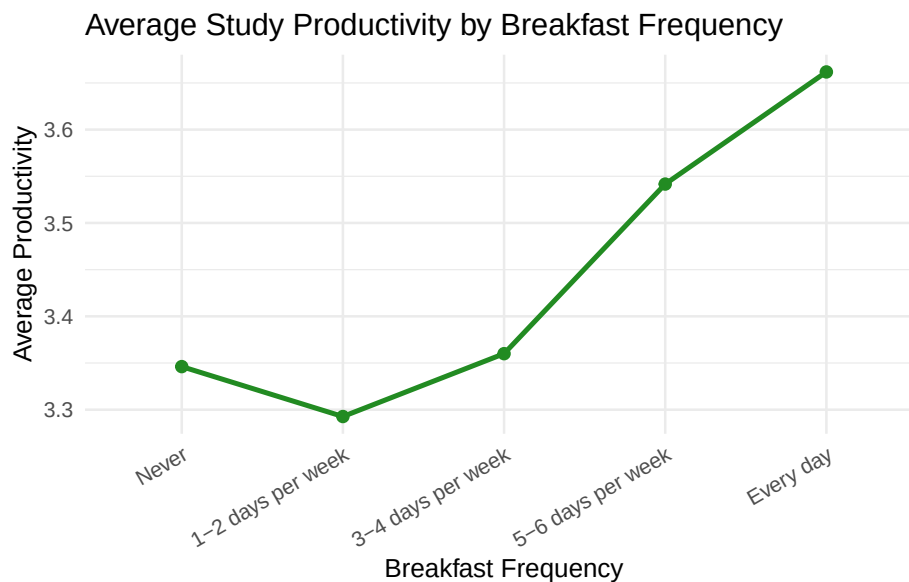
```
h1_summary <- group39_data %>%
  group_by(breakfast_frequency) %>%
  summarise(
    mean_productivity = mean(study_productivity)
  )

ggplot(
  h1_summary,
  aes(
```

```

    x = breakfast_frequency,
    y = mean_productivity,
    group = 1
  )
) +
  geom_line(color = "forestgreen", linewidth = 1) +
  geom_point(color = "forestgreen", size = 2) +
  labs(
    title = "Average Study Productivity by Breakfast Frequency",
    x = "Breakfast Frequency",
    y = "Average Productivity"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))

```



The line chart suggests a slight positive trend between breakfast frequency and average perceived study productivity. Average productivity is lowest among students who eat breakfast one to two days per week and highest among students who eat breakfast every day. However, the differences between categories are relatively small, and the plot only shows group averages rather than the full distribution of responses. Therefore, this visualization should be interpreted as initial exploratory evidence. Overall, the observed pattern is broadly consistent with H1.

Descriptive Inference

To further examine the pattern observed in the line chart, summary statistics were calculated for perceived study productivity grouped by breakfast frequency. This allows comparison of productivity scores across the different breakfast-frequency categories.

```

h1_descriptive <- group39_data %>%
  group_by(breakfast_frequency) %>%
  summarise(
    n = n(),

```

```

mean = round(mean(study_productivity, na.rm = TRUE), 2),
sd = round(sd(study_productivity, na.rm = TRUE), 2),
min = min(study_productivity, na.rm = TRUE),
q25 = round(quantile(study_productivity, 0.25, na.rm = TRUE), 2),
median = median(study_productivity, na.rm = TRUE),
q75 = round(quantile(study_productivity, 0.75, na.rm = TRUE), 2),
max = max(study_productivity, na.rm = TRUE),
.groups = "drop"
)

kable(
  h1_descriptive,
  caption = "Summary statistics of perceived study productivity by breakfast frequency",
  col.names = c(
    "Breakfast Frequency",
    "n",
    "Mean Productivity",
    "SD",
    "Min",
    "Q25",
    "Median",
    "Q75",
    "Max"
  ),
  booktabs = TRUE
)

```

Table 1: Summary statistics of perceived study productivity by breakfast frequency

Breakfast Frequency	n	Mean Productivity	SD	Min	Q25	Median	Q75	Max
Never	26	3.35	0.89	2	3	3	4	5
1-2 days per week	41	3.29	0.81	2	3	3	4	5
3-4 days per week	50	3.36	0.69	2	3	3	4	5
5-6 days per week	24	3.54	0.78	2	3	4	4	5
Every day	68	3.66	0.80	2	3	4	4	5

The descriptive statistics show small differences in perceived study productivity across breakfast-frequency categories. Students who eat breakfast every day report the highest average productivity score (3.66), while those who eat breakfast one to two days per week report the lowest average score (3.29). The median productivity score is 3 for the lower breakfast-frequency categories and increases to 4 for students who eat breakfast five to six days per week or every day. However, the differences between the groups are relatively small, and the pattern is not perfectly monotonic because students who never eat breakfast report a slightly higher average productivity score than students who eat breakfast one to two days per week. Overall, the descriptive results suggest a weak positive pattern and are broadly consistent with H1.

Analytic Inference

To formally test the hypothesis, we use Spearman's rank correlation coefficient (ρ) as the test statistic, because both variables are ordinal. Spearman's correlation evaluates monotonic relationships based on ranks and does not require normally distributed variables.

```

h1_correlation <- cor.test(
  group39_data$breakfast_score,
  group39_data$study_productivity,
  method = "spearman",
  use = "complete.obs"
)

cat(
  "  SPEARMAN CORRELATION TEST RESULTS - H1\n",
  "Testing: Breakfast Frequency vs Study Productivity\n\n",
  sprintf(
    "Correlation coefficient (rho): %.3f\n",
    h1_correlation$estimate
  ),
  sprintf(
    "p-value: %.4f\n",
    h1_correlation$p.value
  ),
  sprintf(
    "Sample size (n): %d\n",
    sum(
      complete.cases(
        group39_data$breakfast_score,
        group39_data$study_productivity
      )
    )
  ),
  sep = ""
)

```

```

##  SPEARMAN CORRELATION TEST RESULTS - H1
## Testing: Breakfast Frequency vs Study Productivity
##
## Correlation coefficient (rho): 0.197
## p-value: 0.0042
## Sample size (n): 209

```

The Spearman correlation test revealed a weak positive correlation between breakfast frequency and perceived study productivity ($\rho = 0.197$). The p-value was $p = 0.0042$, which is below the significance level of $\alpha = 0.05$. Therefore, the null hypothesis of no association can be rejected.

However, the correlation coefficient indicates that the association is weak. This means that students who eat breakfast more frequently tend to report slightly higher perceived study productivity, but breakfast frequency explains only a limited part of the variation in productivity scores. Overall, Hypothesis 1 is supported by the data, but the strength of the relationship is small.

Hypothesis 2: Main Meals per Day and Study Productivity

Research Question: Is the number of main meals per day associated with students' perceived study productivity?

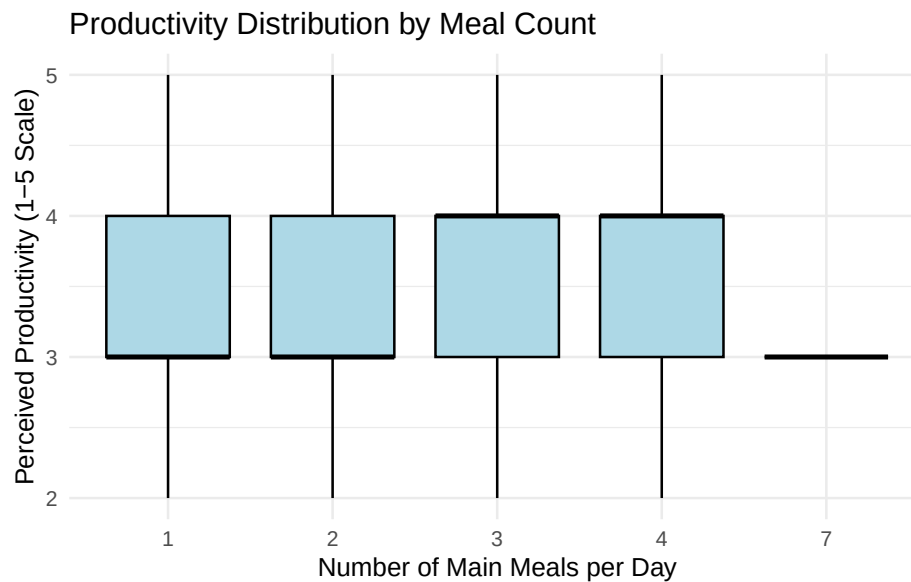
Hypothesis: Students who eat more main meals per day report higher perceived study productivity.

To test this hypothesis, we examine the relationship between two variables: `main_meals_per_day` and `study_productivity`. The variable `main_meals_per_day` is a discrete quantitative variable, as it represents the reported number of main meals students usually eat per day. The variable `study_productivity` is an ordinal variable measured on a 1-5 Likert scale, where higher values indicate higher perceived study productivity.

Exploratory Data Analysis

For the exploratory data analysis, we generate boxplots to visualize the distribution of perceived study productivity scores across the different numbers of reported main meals per day.

```
ggplot(group39_data, aes(x = factor(main_meals_per_day), y = study_productivity)) +  
  geom_boxplot(fill = "lightblue", color = "black") +  
  labs(  
    title = "Productivity Distribution by Meal Count",  
    x = "Number of Main Meals per Day",  
    y = "Perceived Productivity (1-5 Scale)"  
  ) +  
  theme_minimal()
```



The boxplot shows that the distributions of perceived study productivity are very similar across students reporting one to four main meals per day. Most productivity scores are between 3 and 4, and the interquartile ranges overlap strongly across these meal-count groups. This suggests that there is no clear visual evidence of a strong association between the number of main meals per day and perceived study productivity.

The category of seven main meals per day was kept in the visualization to reflect the complete dataset. However, this category appears to be an outlier and should be interpreted with caution, as it contains no visible spread in productivity values.

Descriptive Inference

To further examine the pattern observed in the boxplot, we calculate summary statistics for perceived study productivity grouped by the number of main meals per day. This allows us to compare productivity scores across the different meal-count groups.


```

h2_descriptive <- group39_data %>%
  group_by(main_meals_per_day) %>%
  summarise(
    n = n(),
    mean = round(mean(study_productivity, na.rm = TRUE), 2),
    sd = round(sd(study_productivity, na.rm = TRUE), 2),
    min = min(study_productivity, na.rm = TRUE),
    q25 = round(quantile(study_productivity, 0.25, na.rm = TRUE), 2),
    median = median(study_productivity, na.rm = TRUE),
    q75 = round(quantile(study_productivity, 0.75, na.rm = TRUE), 2),
    max = max(study_productivity, na.rm = TRUE),
    .groups = "drop"
  )

kable(
  h2_descriptive,
  caption = "Summary statistics of perceived study productivity by number of main meals per day",
  col.names = c("Meals", "n", "Mean Productivity", "SD", "Min", "Q25", "Median", "Q75", "Max"),
  booktabs = TRUE
)

```

Table 2: Summary statistics of perceived study productivity by number of main meals per day

Meals	n	Mean Productivity	SD	Min	Q25	Median	Q75	Max
1	17	3.47	1.01	2	3	3	4	5
2	95	3.39	0.72	2	3	3	4	5
3	82	3.52	0.85	2	3	4	4	5
4	14	3.64	0.74	2	3	4	4	5
7	1	3.00	NA	3	3	3	3	3

The descriptive statistics confirm the visual impression from the boxplot. The category of seven main meals per day contains only one observation. Since the category of seven main meals per day contains only one observation and is far outside the range of the other reported meal counts, it was excluded from the inferential test of H2 to avoid giving a single case disproportionate influence. However, it is still shown in the descriptive table to transparently reflect the complete dataset.

Focusing on students reporting one to four main meals per day, perceived study productivity is generally concentrated around values 3 and 4. Students reporting one or two main meals per day have a median productivity score of 3, while students reporting three or four main meals per day have a median productivity score of 4.

The mean productivity scores show a slight increase from 3.47 for students reporting one main meal per day to 3.64 for students reporting four main meals per day. However, the differences between the groups are small, and the quartiles overlap strongly. Therefore, the descriptive statistics suggest at most a weak positive pattern between the number of main meals per day and perceived study productivity.

Analytic Inference

To formally test the hypothesis, we use Spearman's rank correlation coefficient (ρ) as the test statistic. This test is appropriate because perceived study productivity is measured on an ordinal 1-5 Likert scale, while the number of main meals per day is a discrete quantitative variable. Standard parametric correlation

tests, such as Pearson's correlation, assume continuous variables, normally distributed data, and a linear relationship. These assumptions are not fully appropriate here because a productivity score of 2 and 3 is not necessarily the same distance apart as a productivity score of 4 and 5. Spearman's correlation is more suitable in this case because it is based on ranks and tests whether there is a monotonic relationship between the two variables.

Since the category of seven main meals per day contains only one observation and was identified as an outlying category in the descriptive analysis, it is excluded from the inferential test.

```
h2_data <- group39_data %>%
  filter(main_meals_per_day != 7)

spearman_res <- cor.test(
  h2_data$main_meals_per_day,
  h2_data$study_productivity,
  method = "spearman",
  exact = FALSE
)

cat(sprintf("Spearman correlation coefficient: %.3f\n", spearman_res$estimate))

## Spearman correlation coefficient: 0.091
```

```
cat(sprintf("p-value: %.4f\n\n", spearman_res$p.value))
```

```
## p-value: 0.1888
```

The Spearman correlation test shows a very small positive correlation between the number of main meals per day and perceived study productivity, $\rho = 0.091$. However, the p-value is $p = 0.1888$, which is higher than the conventional significance level of $\alpha = 0.05$. Therefore, we fail to reject the null hypothesis. This means that the data do not provide sufficient evidence for a statistically significant association between the number of main meals per day and perceived study productivity.

Overall, Hypothesis 2 is not supported by the data.

Hypothesis 3: Breakfast Frequency and Main Meals per Day

Research Question: Is the frequency of eating breakfast associated with the number of main meals students usually eat per day?

Hypothesis: Students who eat breakfast more often also report eating more main meals per day.

To test this hypothesis, we examine the relationship between two variables: `breakfast_frequency` and `main_meals_per_day`. The variable `breakfast_frequency` is an ordinal variable, as its response categories range from “Never” to “Every day” in a meaningful order. The variable `main_meals_per_day` is a discrete quantitative variable, as it represents the reported number of main meals students usually eat per day.

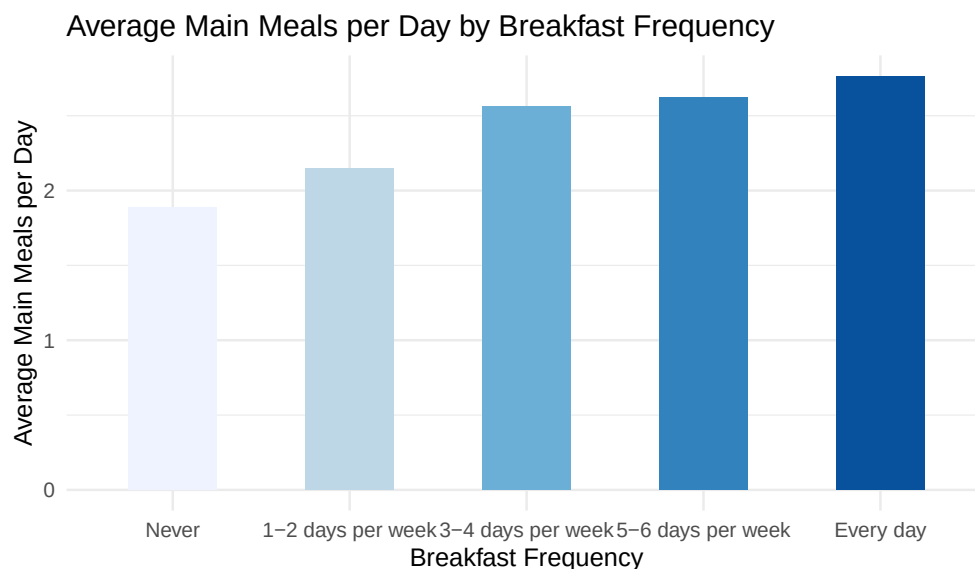
Exploratory Data Analysis

To explore the relationship between breakfast frequency and the number of main meals per day, we calculated the average number of main meals per day for each breakfast frequency category using the `aggregate()` function. This allows us to observe the central tendency of meal consumption across different breakfast

habits. A barplot was created to visualize this relationship, with breakfast frequency on the x-axis and average main meals per day on the y-axis. This visualization helps identify potential trends and patterns in the data.

```
# Calculate average main meals for each breakfast category
h3_summary <- aggregate(
  main_meals_per_day ~ breakfast_frequency,
  data = group39_data,
  FUN = mean
)

# Create barplot for Hypothesis 3
ggplot(h3_summary, aes(x = breakfast_frequency,
  y = main_meals_per_day, fill = breakfast_frequency)) +
  geom_bar(stat = "identity", width = 0.5) +
  labs(
    title = "Average Main Meals per Day by Breakfast Frequency",
    x = "Breakfast Frequency",
    y = "Average Main Meals per Day",
    fill = "Breakfast Frequency"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
  ) +
  scale_fill_brewer(palette = "Blues")
```



The exploratory analysis reveals a positive trend between breakfast frequency and the number of main meals consumed per day. Students who never eat breakfast appear to report the lowest average number of main meals per day, while students who eat breakfast every day appear to report the highest average number. The barplot shows a consistent upward pattern: as breakfast frequency increases from “Never” to “Every day”, the average number of main meals per day also increases. This initial visual evidence is consistent with the hypothesis that students who eat breakfast more frequently also tend to consume more main meals throughout the day.

Descriptive Inference

To compute the descriptive statistics, we grouped the data by breakfast frequency and calculated the count, mean, standard deviation, minimum, first quartile, median, third quartile, and maximum for the number of main meals per day. This provides an overview of both the central tendency and the dispersion of meal consumption across the different breakfast frequency categories.

```
# Compute comprehensive summary statistics for main meals per day by breakfast frequency
h3_descriptive <- group39_data %>%
  group_by(breakfast_frequency) %>%
  summarise(
    n = n(),
    mean = round(mean(main_meals_per_day, na.rm = TRUE), 2),
    sd = round(sd(main_meals_per_day, na.rm = TRUE), 2),
    min = min(main_meals_per_day, na.rm = TRUE),
    q25 = round(quantile(main_meals_per_day, 0.25, na.rm = TRUE), 2),
    median = median(main_meals_per_day, na.rm = TRUE),
    q75 = round(quantile(main_meals_per_day, 0.75, na.rm = TRUE), 2),
    max = max(main_meals_per_day, na.rm = TRUE),
    .groups = "drop"
  )

# Display the table
kable(
  h3_descriptive,
  caption = "Summary statistics of main meals per day by breakfast frequency",
  col.names = c("Breakfast Frequency", "n", "Mean", "SD", "Min", "Q25", "Median", "Q75", "Max"),
  booktabs = TRUE
)
```

Table 3: Summary statistics of main meals per day by breakfast frequency

Breakfast Frequency	n	Mean	SD	Min	Q25	Median	Q75	Max
Never	26	1.88	0.65	1	1.25	2	2	3
1-2 days per week	41	2.15	0.48	1	2.00	2	2	3
3-4 days per week	50	2.56	0.95	1	2.00	2	3	7
5-6 days per week	24	2.62	0.92	1	2.00	3	3	4
Every day	68	2.76	0.67	1	2.00	3	3	4

The descriptive statistics confirm the visual impression from the barplot. Students who never eat breakfast report the lowest average number of main meals per day, while students who eat breakfast every day report the highest average number. The mean number of main meals increases consistently across the breakfast frequency categories, from 1.88 among students who never eat breakfast to 2.76 among students who eat breakfast every day.

The median number of main meals is 2 in the lower breakfast-frequency categories and increases to 3 among students who eat breakfast five to six days per week or every day. Although there is some variability within the groups, especially in the “3-4 days per week” and “5-6 days per week” categories, the overall descriptive pattern suggests a positive association between breakfast frequency and the number of main meals per day. Therefore, the descriptive statistics are consistent with H3.

Analytic Inference

To test whether there is a statistically significant association between breakfast frequency and the number of main meals per day, we used Spearman's rank correlation coefficient. This non-parametric test is appropriate because breakfast frequency is an ordinal variable with ordered response categories, and the aim is to assess whether there is a monotonic relationship between breakfast frequency and the number of main meals per day without assuming normally distributed variables.

```
# Spearman correlation test
h3_correlation <- cor.test(
  group39_data$breakfast_score,
  group39_data$main_meals_per_day,
  method = "spearman",
  use = "complete.obs"
)

# Display results
cat(
  "  SPEARMAN CORRELATION TEST RESULTS - H3\n",
  "Testing: Breakfast Frequency vs Main Meals per Day\n\n",
  sprintf("Correlation coefficient (rho):  %.3f\n", h3_correlation$estimate),
  sprintf("p-value:                        %.4f\n", h3_correlation$p.value),
  sprintf(
    "Sample size (n):                      %d\n",
    sum(complete.cases(group39_data$breakfast_score, group39_data$main_meals_per_day))
  ),
  sep = ""
)
```

```
##  SPEARMAN CORRELATION TEST RESULTS - H3
## Testing: Breakfast Frequency vs Main Meals per Day
##
## Correlation coefficient (rho):  0.425
## p-value:                        0.0000
## Sample size (n):                209
```

The Spearman correlation test shows a moderate positive correlation between breakfast frequency and the number of main meals per day, $\rho = 0.425$. The p-value is smaller than 0.001, which is below the conventional significance level of $\alpha = 0.05$. Therefore, we reject the null hypothesis of no association. This means that the data provide evidence for a statistically significant positive association between breakfast frequency and the number of main meals per day. As breakfast frequency increases, the number of main meals per day also tends to increase.

Overall, Hypothesis 3 is supported by the data. Students who eat breakfast more frequently tend to report eating more main meals per day.

Additional Exploratory Analysis

In addition to the main hypothesis tests, we explore whether demographic variables (age, gender, and academic program) provide additional context for the relationships between meal habits and perceived study productivity. These analyses are intended as additional exploratory context and we visually inspect whether the observed patterns appear to differ across demographic groups.

Before interpreting subgroup patterns, we first inspect the distribution of the demographic variables. This helps assess whether some groups are strongly underrepresented and should therefore be interpreted with caution.

```
demographic_counts <- bind_rows(
  group39_data %>%
    count(gender, name = "n") %>%
    mutate(Variable = "Gender", Category = as.character(gender)) %>%
    select(Variable, Category, n),

  group39_data %>%
    count(age, name = "n") %>%
    mutate(Variable = "Age", Category = as.character(age)) %>%
    select(Variable, Category, n),

  group39_data %>%
    count(academic_program, name = "n") %>%
    mutate(Variable = "Academic Program", Category = as.character(academic_program)) %>%
    select(Variable, Category, n)
)

kable(
  demographic_counts,
  caption = "Distribution of demographic variables",
  col.names = c("Variable", "Category", "n"),
  booktabs = TRUE
)
```

Table 4: Distribution of demographic variables

Variable	Category	n
Gender	Female	53
Gender	Male	156
Age	18-20	3
Age	21-23	30
Age	23-25	55
Age	25-30	98
Age	30-35	15
Age	over 35	8
Academic Program	Business Informatics	32
Academic Program	Data Science	177

The table shows that the sample is unevenly distributed across demographic groups. In particular, most participants are male, between 25 and 30 years old, and enrolled in Data Science. Therefore, the following subgroup analyses are interpreted cautiously and used only as additional exploratory context.

Demographic Differences in the Meal-Productivity Relationship

As an additional exploratory analysis, we examine whether the relationship between main meals per day and perceived study productivity differs across demographic groups. This analysis is not part of the main hypothesis testing framework, but provides additional context for interpreting the main results.

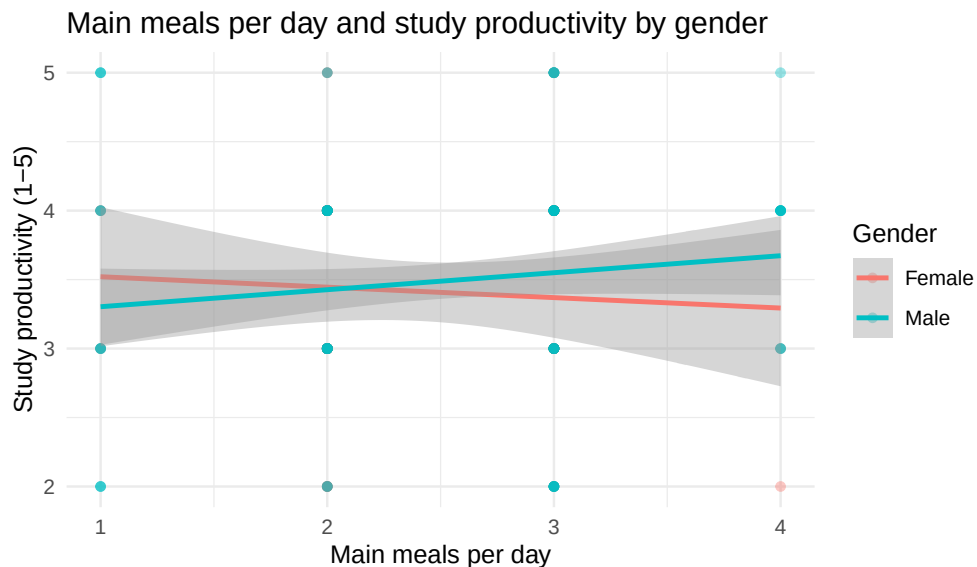
Gender

We first explore whether the association between main meals per day and perceived study productivity differs by gender.

```
group39_data <- group39_data %>%  
  filter(main_meals_per_day != 7)
```

We again exclude the person with seven meals per day from further analysis.

```
ggplot(group39_data, aes(x = main_meals_per_day, y = study_productivity, color = gender)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = TRUE) +  
  labs(  
    x = "Main meals per day",  
    y = "Study productivity (1-5)",  
    title = "Main meals per day and study productivity by gender",  
    color = "Gender"  
  ) +  
  theme_minimal()
```



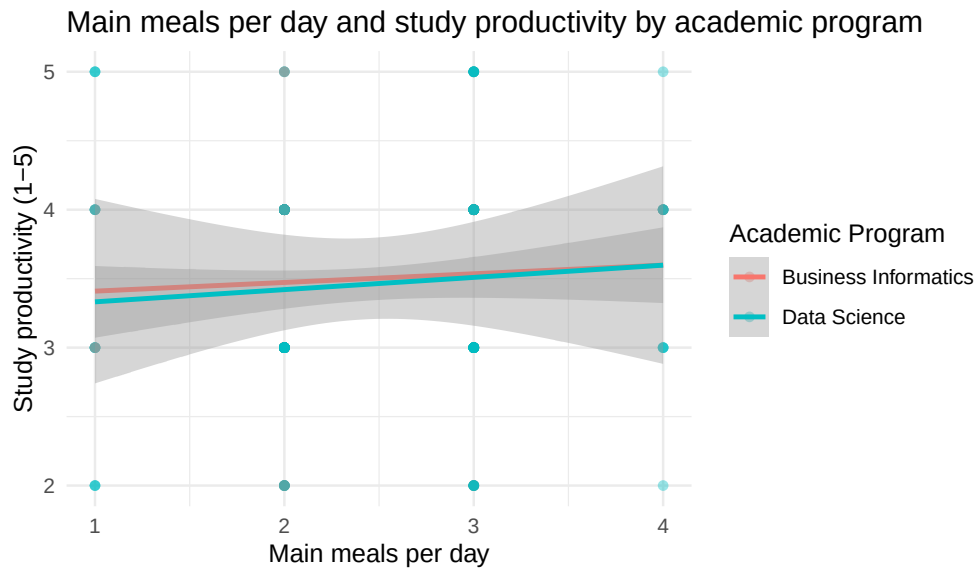
The plot suggests a possible gender difference in the relationship between main meals per day and perceived study productivity. While the trend appears slightly positive for male students, it appears slightly negative for female students. However, no strong conclusion should be drawn from this visual pattern alone.

Academic Program

We also explore whether the association between main meals per day and perceived study productivity differs by academic program.

```
ggplot(group39_data, aes(x = main_meals_per_day, y = study_productivity, color = academic_program)) +  
  geom_point(alpha = 0.4) +
```

```
geom_smooth(method = "lm", se = TRUE) +
labs(
  x = "Main meals per day",
  y = "Study productivity (1-5)",
  title = "Main meals per day and study productivity by academic program",
  color = "Academic Program"
) +
theme_minimal()
```

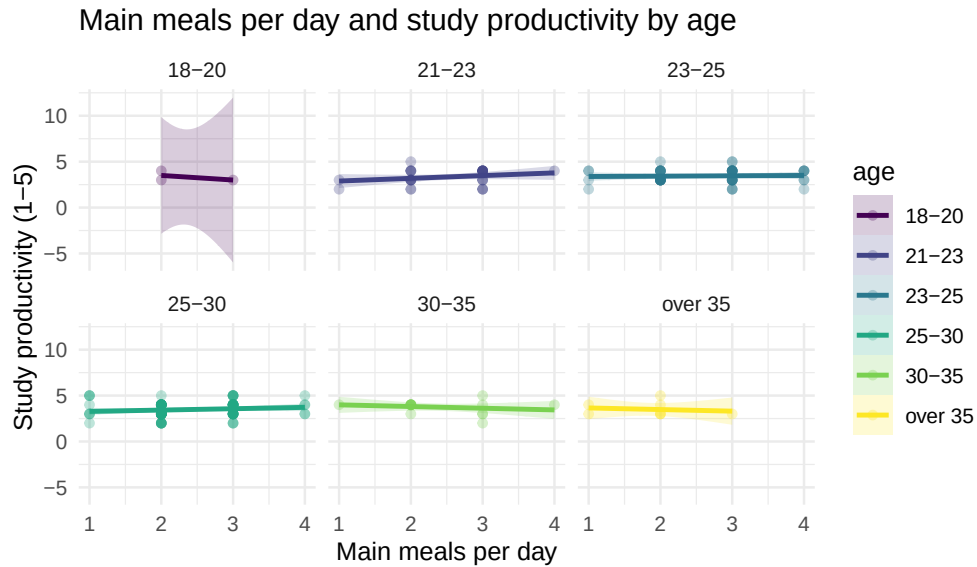


For academic program, the regression lines are very similar and point in roughly the same direction. This suggests that academic program does not visibly change the relationship between main meals per day and perceived study productivity in a meaningful way. Therefore, this variable does not appear to add much explanatory value in the exploratory analysis.

Age

Finally, we inspect perceived study productivity across age groups. Because some age groups contain only a small number of observations, this comparison should be interpreted cautiously.

```
ggplot(group39_data, aes(x = main_meals_per_day, y = study_productivity, color = age)) +
  geom_point(alpha = 0.3) +
  facet_wrap(~ age) +
  geom_smooth(method = "lm", se = TRUE, aes(fill = age), alpha = 0.2) +
  labs(
    x = "Main meals per day",
    y = "Study productivity (1-5)",
    title = "Main meals per day and study productivity by age"
  ) +
  theme_minimal()
```

The plot shows some variation across age groups, with the relationship between main meals per day and perceived study productivity moving slightly up and down rather than following a clear overall trend. Because several age groups contain only a few observations, these age-specific patterns should be interpreted cautiously and only as exploratory context.

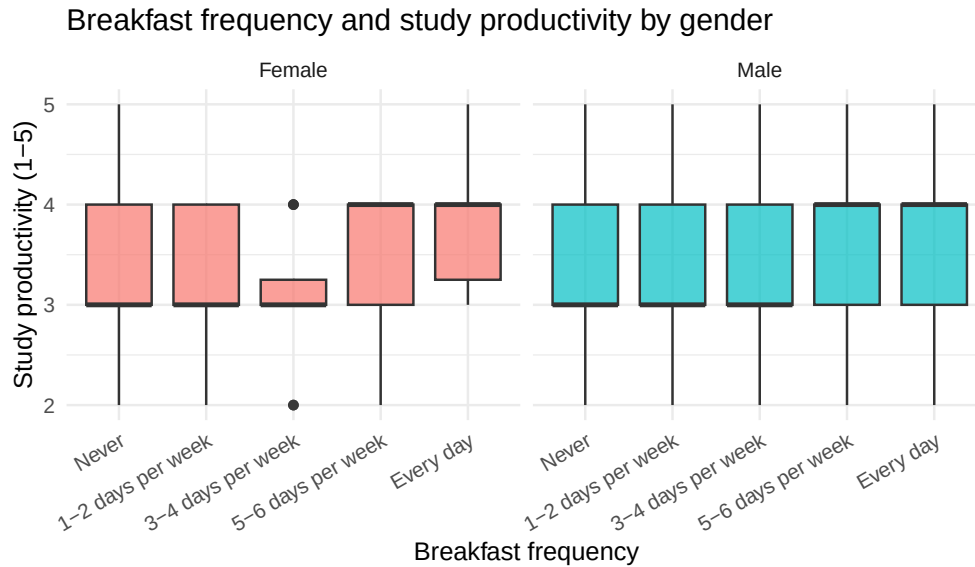
Demographic Differences in the Breakfast-Productivity Relationship

As an additional exploratory analysis, we also briefly examine whether the relationship between breakfast frequency per week and perceived study productivity differs across demographic groups. For the breakfast-productivity relationship, we focus on gender and age, as these provide the most interpretable subgroup comparisons.

Gender

We first explore whether the association between breakfast frequency and perceived study productivity differs by gender.

```
ggplot(group39_data, aes(x = breakfast_frequency, y = study_productivity, fill = gender)) +
  geom_boxplot(alpha = 0.7) +
  facet_wrap(~ gender) +
  labs(
    x = "Breakfast frequency",
    y = "Study productivity (1-5)",
    title = "Breakfast frequency and study productivity by gender"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1),
        legend.position = "none")
```



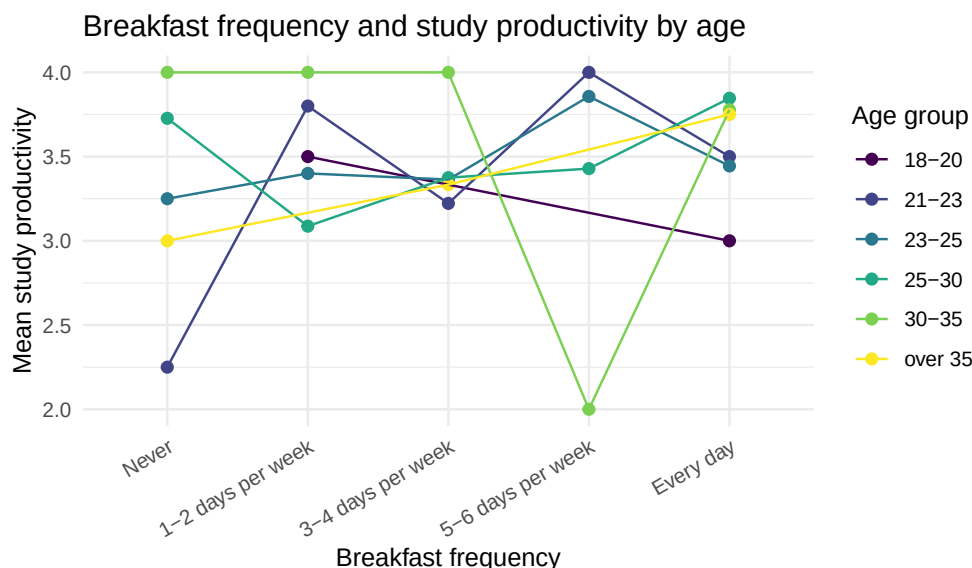
The gender-specific plot does not suggest a strong difference between female and male students. Across breakfast frequency categories, the distribution of perceived study productivity appears broadly similar for both groups. Therefore, gender does not seem to meaningfully change the breakfast-productivity relationship in this sample.

Age

Finally, we inspect perceived study productivity across age groups. Because some age groups contain only a small number of observations, this comparison should be interpreted cautiously.

```
breakfast_age_summary <- group39_data %>%
  group_by(age, breakfast_frequency) %>%
  summarise(
    mean_productivity = mean(study_productivity),
    n = n(),
    .groups = "drop"
  )

ggplot(breakfast_age_summary, aes(
  x = breakfast_frequency,
  y = mean_productivity,
  group = age,
  color = age
)) +
  geom_point(size = 2) +
  geom_line() +
  labs(
    x = "Breakfast frequency",
    y = "Mean study productivity",
    title = "Breakfast frequency and study productivity by age",
    color = "Age group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



The age-specific plot shows more variation across groups. Some age groups show slightly higher average productivity among students who eat breakfast every day, but the pattern is not fully consistent across all age categories. Age may be an interesting factor for further exploratory analysis, but some age-by-breakfast categories may contain very few observations, so individual mean values should not be overinterpreted.

Summary of Findings and Conclusion

This report examined the relationship between meal habits and perceived study productivity among university students. The analysis focused on three research questions concerning breakfast frequency, the number of main meals per day, and perceived study productivity. The analysis uses `read.csv()` to import the data, `mutate()` to transform variables, `factor()` to define ordered categorical variables, `ggplot()` to create visualizations, `group_by()` and `summarise()` to compute descriptive statistics, `kable()` to display summary tables, and `cor.test()` to conduct Spearman correlation tests.

For Hypothesis 1, the relationship between breakfast frequency and perceived study productivity was examined. The results showed a weak but statistically significant positive association between breakfast frequency and perceived study productivity. Students who reported eating breakfast more frequently tended to report slightly higher levels of perceived study productivity. However, the correlation coefficient was small, indicating that the strength of this relationship is limited. Therefore, Hypothesis 1 was supported by the data. For Hypothesis 2, the results did not provide evidence for a statistically significant association between the number of main meals per day and perceived study productivity. Although the descriptive statistics showed a slight positive pattern, the Spearman correlation was very small and not statistically significant. Therefore, Hypothesis 2 was not supported by the data. For Hypothesis 3, the results showed a statistically significant moderate positive association between breakfast frequency and the number of main meals per day. Students who reported eating breakfast more frequently also tended to report eating more main meals per day. Therefore, Hypothesis 3 was supported by the data.

The additional exploratory analysis provided further demographic context. Some visual subgroup patterns appeared potentially interesting, for example differences by gender or age. However, these analyses were not part of the main hypothesis testing framework and should not be interpreted as evidence for additional hypotheses. Since some demographic subgroups were relatively small, these patterns should be considered exploratory only and could be investigated more systematically in future research.

Overall, the findings suggest that breakfast frequency is clearly related to overall meal frequency, while the connection between meal habits and perceived study productivity is less clear. The results therefore indicate

that eating patterns may be associated with each other, but they do not provide strong evidence that eating more main meals per day is linked to higher perceived study productivity in this sample.